

Errata and comments on JÜL-2891

M. Baelmans

December 28, 2012

- p.25, Figure 2.1 The direction of the basis vector $\vec{e}_\perp$ must be reversed
- p30, Eq. (2.64)

$$\begin{aligned} \chi_{div,\parallel} = & \frac{\tilde{B}_\theta}{\tilde{B}} mn \tilde{V}_r \tilde{V}_\theta \frac{\partial \ln h_\theta}{h_r \partial u^r} - \frac{\tilde{B}_\theta}{\tilde{B}} mn \tilde{V}_r^2 \frac{\partial \ln h_r}{h_\theta \partial u^\theta} \\ & + \frac{\tilde{B}_\phi}{\tilde{B}} mn \tilde{V}_r \tilde{V}_\phi \frac{\partial \ln h_\phi}{h_r \partial u^r} + mn \tilde{V}_\perp \tilde{V}_\phi \frac{\partial \ln h_\phi}{h_\theta \partial u^\theta} . \end{aligned} \quad (1)$$

instead of :

$$\begin{aligned} \chi_{div,\parallel} = & \frac{\tilde{B}_\theta}{\tilde{B}} mn \tilde{V}_r \tilde{V}_\theta \frac{\partial \ln h_\theta}{h_r \partial u^r} - \frac{\tilde{B}_\theta}{\tilde{B}} mn \tilde{V}_r^2 \frac{\partial \ln h_r}{h_\theta \partial u^\theta} \\ & + \frac{\tilde{B}_\phi}{\tilde{B}} mn \tilde{V}_r \tilde{V}_\phi \frac{\partial \ln h_\phi}{h_\theta \partial u^\theta} + mn \tilde{V}_\perp \tilde{V}_\phi \frac{\partial \ln h_\phi}{h_\theta \partial u^\theta} . \end{aligned} \quad (2)$$

- p. 32, Eq. (2.73)

$$\begin{aligned} \beta^* = & -\frac{\tilde{B}_\theta}{\tilde{B}} \frac{\tilde{V}_\theta}{h_\theta} \frac{\partial \tilde{B}_\theta / \tilde{B}}{\partial u^\theta} - \frac{\tilde{B}_\phi}{\tilde{B}} \frac{\tilde{V}_\phi}{h_\theta} \frac{\partial \tilde{B}_\phi / \tilde{B}}{\partial u^\theta} \\ & - \frac{1}{3} \tilde{V}_\parallel \frac{\partial \tilde{B}_\theta / \tilde{B}}{h_\theta \partial u^\theta} - \frac{1}{3} \tilde{V}_\perp \frac{\partial \tilde{B}_\phi / \tilde{B}}{h_\theta \partial u^\theta} \end{aligned} \quad (3)$$

instead of :

$$\begin{aligned} \beta^* = & -\frac{\tilde{B}_\theta}{\tilde{B}} \frac{\tilde{V}_\theta}{h_\theta} \frac{\partial \tilde{B}_\theta / \tilde{B}}{\partial u^\theta} - \frac{\tilde{B}_\phi}{\tilde{B}} \frac{\tilde{V}_\phi}{h_\theta} \frac{\partial \tilde{B}_\phi / \tilde{B}}{\partial u^\theta} \\ & - \frac{1}{3} \tilde{V}_\parallel \frac{\partial \tilde{B}_\theta / \tilde{B}}{h_\theta \partial u^\theta} - \frac{1}{3} \tilde{V}_\perp \frac{\partial \tilde{B}_\phi / \tilde{B}}{h_\theta \partial u^\theta} \end{aligned} \quad (4)$$

- p. 32, Eq. (2.74)

$$\begin{aligned}\chi^* = & -\frac{1}{3}\tilde{V}_\theta \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^\theta} \left(\frac{\sqrt{g}}{h_\theta} \right) - \frac{1}{3}\tilde{V}_r \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^r} \left(\frac{\sqrt{g}}{h_r} \right) \\ & + \tilde{V}_r \left(\frac{\tilde{B}_\theta^2}{\tilde{B}^2} \frac{1}{h_r} \frac{\partial \ln h_\theta}{\partial u^r} + \frac{\tilde{B}_\phi^2}{\tilde{B}^2} \frac{1}{h_r} \frac{\partial \ln h_\phi}{\partial u^r} \right) - \tilde{V}_\perp \frac{\tilde{B}_\phi}{\tilde{B}} \frac{1}{h_\theta} \frac{\partial \ln h_\phi}{\partial u^\theta} \quad .\end{aligned}\tag{5}$$

instead of

$$\begin{aligned}\chi^* = & -\frac{1}{3}\tilde{V}_\theta \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^\theta} \left(\frac{\sqrt{g}}{h_\theta} \right) - \frac{1}{3}\tilde{V}_r \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^r} \left(\frac{\sqrt{g}}{h_r} \right) \\ & + \tilde{V}_r \left(\frac{\tilde{B}_\theta^2}{\tilde{B}^2} \frac{1}{h_r} \frac{\partial \ln h_\theta}{\partial u^r} + \frac{\tilde{B}_\phi^2}{\tilde{B}^2} \frac{1}{h_r} \frac{\partial \ln h_\phi}{\partial u^r} \right) - \tilde{V}_\phi \frac{\tilde{B}_\theta \tilde{B}_\phi}{\tilde{B}^2} \frac{1}{h_\theta} \frac{\partial \ln h_\phi}{\partial u^\theta} \quad .\end{aligned}\tag{6}$$

- p. 32, Eq. (2.75)

$$\beta^* = \left(-\tilde{V}_\parallel + \frac{2}{3}\tilde{V}_\phi \frac{\tilde{B}}{\tilde{B}_\phi} \right) \frac{\partial \tilde{B}_\theta / \tilde{B}}{h_\theta \partial u^\theta} \quad .\tag{7}$$

instead of

$$\beta^* = -\tilde{V}_\parallel + \tilde{V}_\phi \left(3 \frac{\tilde{B}_\phi}{\tilde{B}} + 3 \frac{\tilde{B}_\theta}{\tilde{B}} - \frac{\tilde{B}}{3\tilde{B}_\phi} \right) \frac{\partial \tilde{B}_\theta / \tilde{B}}{h_\theta \partial u^\theta} \quad .\tag{8}$$

- p. 32, Eq. (2.77)

$$\begin{aligned}\tilde{\Pi}_{\theta\theta} &= -3\eta_0 \left(\frac{\tilde{B}_\theta^2}{\tilde{B}^2} - \frac{1}{3} \right) \tilde{W}_{\parallel\parallel} \\ \tilde{\Pi}_{\phi\phi} &= -3\eta_0 \left(\frac{\tilde{B}_\phi^2}{\tilde{B}^2} - \frac{1}{3} \right) \tilde{W}_{\parallel\parallel} \\ \tilde{\Pi}_{\phi\theta} &= \tilde{\Pi}_{\theta\phi} = -3\eta_0 \frac{\tilde{B}_\theta \tilde{B}_\phi}{\tilde{B}^2} \tilde{W}_{\parallel\parallel} \\ \tilde{\Pi}_{rr} &= \eta_0 \tilde{W}_{\parallel\parallel} \\ \tilde{\Pi}_{r\theta} &= \tilde{\Pi}_{r\phi} = \tilde{\Pi}_{\theta r} = \tilde{\Pi}_{\phi r} = 0 \quad .\end{aligned}\tag{9}$$

instead of

$$\begin{aligned}\tilde{\Pi}_{\theta\theta} &= -3\eta_0 \left(\frac{\tilde{B}_\theta^2}{\tilde{B}^2} - \frac{1}{3} \right) \tilde{W}_{\parallel\parallel} \\ \tilde{\Pi}_{\phi\phi} &= -3\eta_0 \left(\frac{\tilde{B}_\phi^2}{\tilde{B}^2} - \frac{1}{3} \right) \tilde{W}_{\parallel\parallel} \\ \tilde{\Pi}_{\phi\theta} &= \tilde{\Pi}_{\theta\phi} = -3\eta_0 \left(\frac{\tilde{B}_\theta \tilde{B}_\phi}{\tilde{B}^2} - \frac{1}{3} \right) \tilde{W}_{\parallel\parallel} \\ \tilde{\Pi}_{rr} &= \eta_0 \tilde{W}_{\parallel\parallel} \\ \tilde{\Pi}_{r\theta} &= \tilde{\Pi}_{r\phi} = \tilde{\Pi}_{\theta r} = \tilde{\Pi}_{\phi r} = 0 \quad .\end{aligned}\tag{10}$$

- p. 33, Eq. (2.85)

$$\begin{aligned}
\beta_{\Pi,\parallel} = & \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^\theta} \left\{ \frac{\sqrt{g}}{h_\theta} 2\eta_0 \frac{\tilde{B}_\theta}{\tilde{B}} \left[\tilde{V}_\parallel - \tilde{V}_\phi \left(\frac{2}{3} \frac{\tilde{B}_\phi}{\tilde{B}} \right) \right] \frac{\partial \tilde{B}_\theta / \tilde{B}}{h_\theta \partial u^\theta} \right\} \\
& + \eta_0 \tilde{W}_{\parallel\parallel} \left[-\frac{1}{h_\theta} \frac{\partial}{\partial u^\theta} \left(\frac{\tilde{B}_\theta}{\tilde{B}} \right) \right]
\end{aligned} \tag{11}$$

instead of :

$$\begin{aligned}
\beta_{\Pi,\parallel} = & \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^\theta} \left\{ \frac{\sqrt{g}}{h_\theta} 2\eta_0 \frac{\tilde{B}_\theta}{\tilde{B}} \left[\tilde{V}_\parallel + \tilde{V}_\phi \left(-\frac{\tilde{B}_\phi}{\tilde{B}} - \frac{\tilde{B}_\theta}{\tilde{B}} + \frac{\tilde{B}}{3\tilde{B}_\phi} \right) \right] \frac{\partial \tilde{B}_\theta / \tilde{B}}{h_\theta \partial u^\theta} \right\} \\
& + \eta_0 \tilde{W}_{\parallel\parallel} \left[\left(4 \frac{\tilde{B}_\theta^2}{\tilde{B}^2} - 1 \right) \frac{1}{h_\theta} \frac{\partial}{\partial u^\theta} \left(\frac{\tilde{B}_\theta}{\tilde{B}} \right) \right]
\end{aligned} \tag{12}$$

- p. 34, Eq. (2.88)

$$\begin{aligned}
\beta_{\Pi,\perp} = & \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^\theta} \left\{ \frac{\sqrt{g}}{h_\theta} \eta_0 \frac{\tilde{B}_\phi}{\tilde{B}} \left(-\tilde{V}_\parallel + \tilde{V}_\phi \frac{2}{3} \frac{\tilde{B}_\phi}{\tilde{B}} \right) \frac{1}{h_\theta} \frac{\partial \tilde{B}_\theta / \tilde{B}}{\partial u^\theta} \right\} \\
& - \eta_0 \tilde{W}_{\parallel\parallel} \cdot 2 \frac{\tilde{B}_\theta}{\tilde{B}} \frac{1}{h_\theta} \frac{\partial \tilde{B}_\theta / \tilde{B}}{\partial u^\theta}
\end{aligned} \tag{13}$$

instead of :

$$\begin{aligned}
\beta_{\Pi,\perp} = & \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^\theta} \left\{ \frac{\sqrt{g}}{h_\theta} \eta_0 \frac{\tilde{B}_\phi}{\tilde{B}} \left[-\tilde{V}_\parallel + \tilde{V}_\phi \left(\frac{\tilde{B}_\phi}{\tilde{B}} + \frac{\tilde{B}_\theta}{\tilde{B}} - \frac{\tilde{B}}{3\tilde{B}_\phi} \right) \right] \frac{1}{h_\theta} \frac{\partial \tilde{B}_\theta / \tilde{B}}{\partial u^\theta} \right\} \\
& + \eta_0 \tilde{W}_{\parallel\parallel} \cdot 2 \frac{\tilde{B}_\theta}{\tilde{B}} \left(-\frac{\tilde{B}}{\tilde{B}_\phi} + 3 \frac{\tilde{B}_\phi}{\tilde{B}} \right) \frac{1}{h_\theta} \frac{\partial \tilde{B}_\theta / \tilde{B}}{\partial u^\theta}
\end{aligned} \tag{14}$$

- p. 34, Eq. (2.89)

$$\begin{aligned}
\chi_{\Pi,\perp} = & \\
& -\frac{1}{\sqrt{g}} \frac{\partial}{\partial u^\theta} \left[\frac{\sqrt{g}}{h_\theta} \eta_0 \frac{\tilde{B}_\phi}{\tilde{B}} \left(\frac{1}{3} \tilde{V}_\theta \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^\theta} \frac{\sqrt{g}}{h_\theta} - \frac{1}{3} \tilde{V}_r \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^r} \frac{\sqrt{g}}{h_r} \right) \right. \\
& \left. + \tilde{V}_r \left(\frac{\tilde{B}_\theta^2}{\tilde{B}^2} \frac{1}{h_r} \frac{\partial \ln h_\theta}{\partial u^r} + \frac{\tilde{B}_\phi^2}{\tilde{B}^2} \frac{1}{h_r} \frac{\partial \ln h_\phi}{\partial u^r} \right) \right]
\end{aligned}$$

- p. 35, Eq. (2.92)

$$\begin{aligned}
\chi_{\Pi,r} = & \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^r} \left\{ \frac{\sqrt{g}}{h_r} \eta_0 \left[-\frac{1}{3} \tilde{V}_\theta \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^\theta} \frac{\sqrt{g}}{h_\theta} - \frac{1}{3} \tilde{V}_r \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^r} \frac{\sqrt{g}}{h_r} \right. \right. \\
& + \tilde{V}_r \left(\frac{\tilde{B}_\theta^2}{\tilde{B}^2} \frac{1}{h_r} \frac{\partial \ln h_\theta}{\partial u^r} + \frac{\tilde{B}_\phi^2}{\tilde{B}^2} \frac{1}{h_r} \frac{\partial \ln h_\phi}{\partial u^r} \right) - \tilde{V}_\phi \frac{\tilde{B}_\theta \tilde{B}_\phi}{\tilde{B}^2} \frac{1}{h_\theta} \frac{\partial \ln h_\phi}{\partial u^\theta} \left. \right] \\
& + \eta_0 \tilde{W}_{|||} \left[\left(\frac{3\tilde{B}_\theta^2}{\tilde{B}^2} - 1 \right) \frac{\partial \ln h_\theta}{h_r \partial u^r} + \left(\frac{3\tilde{B}_\phi^2}{\tilde{B}^2} - 1 \right) \frac{\partial \ln h_\phi}{h_r \partial u^r} \right] \Bigg\}
\end{aligned} \tag{19}$$

instead of :

$$\begin{aligned}
\chi_{\Pi,r} = & \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^r} \left\{ \frac{\sqrt{g}}{h_r} \eta_0 \left[-\frac{1}{3} \tilde{V}_\theta \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^\theta} \frac{\sqrt{g}}{h_\theta} - \frac{1}{3} \tilde{V}_r \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^r} \frac{\sqrt{g}}{h_r} \right] \right. \\
& + \tilde{V}_r \left(\frac{\tilde{B}_\theta^2}{\tilde{B}^2} \frac{1}{h_r} \frac{\partial \ln h_\theta}{\partial u^r} + \frac{\tilde{B}_\phi^2}{\tilde{B}^2} \frac{1}{h_r} \frac{\partial \ln h_\phi}{\partial u^r} \right) - \tilde{V}_\phi \frac{\tilde{B}_\theta \tilde{B}_\phi}{\tilde{B}^2} \frac{1}{h_\theta} \frac{\partial \ln h_\phi}{\partial u^\theta} \Bigg\} \\
& + \eta_0 \tilde{W}_{|||} \left[\left(\frac{3\tilde{B}_\theta^2}{\tilde{B}^2} - 1 \right) \frac{\partial \ln h_\theta}{h_r \partial u^r} + \left(\frac{3\tilde{B}_\phi^2}{\tilde{B}^2} - 1 \right) \frac{\partial \ln h_\phi}{h_r \partial u^r} \right]
\end{aligned} \tag{20}$$

- p.36, Eq. (2.98)

$$\tilde{R}_{i,r} = -en_e \frac{\tilde{j}_r}{\sigma_\perp} + \frac{3}{2} \frac{en_e^2 \tilde{B}_\phi}{\sigma_\perp \tilde{B}^2} \frac{1}{h_\theta} \frac{\partial T_e}{\partial u^\theta} \quad . \tag{21}$$

instead of:

$$\tilde{R}_{i,r} = +en_e \frac{\tilde{j}_r}{\sigma_\perp} + \frac{2}{3} \frac{en_e^2 \tilde{B}_\phi}{\sigma_\perp \tilde{B}^2} \frac{1}{h_\theta} \frac{\partial T_e}{\partial u^\theta} \quad . \tag{22}$$

- p.37, Eq. (2.100)

$$\begin{aligned}
& \frac{\partial}{\partial t} \left(m_i n_i \tilde{V}_{i,\perp} \right) + \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^\theta} \left(\frac{\sqrt{g}}{h_\theta} m_i n_i \tilde{V}_{i,\theta} \tilde{V}_{i,\perp} \right) + \\
& \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^r} \left(\frac{\sqrt{g}}{h_r} m_i n_i \tilde{V}_{i,r} \tilde{V}_{i,\perp} \right) + \beta_{div,\perp} + \chi_{div,\perp} + \\
& \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^\theta} \left[\frac{\sqrt{g}}{h_\theta} \eta_0 \frac{\tilde{B}_\phi}{\tilde{B}} \left(\frac{2}{3} \frac{\tilde{B}_\theta}{\tilde{B}} \frac{1}{h_\theta} \frac{\partial \tilde{V}_{i,\parallel}}{\partial u_\theta} - \right. \right. \\
& \left. \left. \frac{1}{3} \frac{\tilde{B}_\phi}{\tilde{B}} \frac{1}{h_\theta} \frac{\partial \tilde{V}_{i,\perp}}{\partial u^\theta} - \frac{1}{3} \frac{\textcolor{green}{1}}{\textcolor{green}{h_r}} \frac{\partial \tilde{V}_{i,r}}{\partial u^r} \right) \right] + \beta_{\Pi,\perp} + \chi_{\Pi,\perp} = \\
& - \frac{\tilde{B}_\phi}{\tilde{B}} \frac{1}{h_\theta} \frac{\partial p_i}{\partial u^\theta} - Z_i e n_i \frac{\tilde{B}_\phi}{\tilde{B}} \frac{1}{h_\theta} \frac{\partial \phi}{\partial u^\theta} \\
& + Z_i e n_i \tilde{V}_r \tilde{B} + \tilde{R}_{i,\perp} + \tilde{S}_{m_i} \vec{V}_{i,\perp}
\end{aligned} \tag{23}$$

instead of:

$$\begin{aligned}
& \frac{\partial}{\partial t} \left(m_i n_i \tilde{V}_{i,\perp} \right) + \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^\theta} \left(\frac{\sqrt{g}}{h_\theta} m_i n_i \tilde{V}_{i,\theta} \tilde{V}_{i,\perp} \right) + \\
& \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^r} \left(\frac{\sqrt{g}}{h_r} m_i n_i \tilde{V}_{i,r} \tilde{V}_{i,\perp} \right) + \beta_{div,\perp} + \chi_{div,\perp} + \\
& \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^\theta} \left[\frac{\sqrt{g}}{h_\theta} \eta_0 \frac{\tilde{B}_\phi}{\tilde{B}} \left(\frac{2}{3} \frac{\tilde{B}_\theta}{\tilde{B}} \frac{1}{h_\theta} \frac{\partial \tilde{V}_{i,\parallel}}{\partial u_\theta} - \right. \right. \\
& \left. \left. \frac{1}{3} \frac{\tilde{B}_\phi}{\tilde{B}} \frac{1}{h_\theta} \frac{\partial \tilde{V}_{i,\perp}}{\partial u^\theta} - \frac{1}{3} \textcolor{red}{h_r} \frac{\partial \tilde{V}_{i,r}}{\partial u^r} \right) \right] + \beta_{\Pi,\perp} + \chi_{\Pi,\perp} = \\
& - \frac{\tilde{B}_\phi}{\tilde{B}} \frac{1}{h_\theta} \frac{\partial p_i}{\partial u^\theta} - Z_i e n_i \frac{\tilde{B}_\phi}{\tilde{B}} \frac{1}{h_\theta} \frac{\partial \phi}{\partial u^\theta} \\
& + Z_i e n_i \tilde{V}_r \tilde{B} + \tilde{R}_{i,\perp} + \tilde{S}_{m_i} \vec{V}_{i,\perp}
\end{aligned} \tag{24}$$

- p.37, Eq. (2.101)

$$\tilde{R}_{i,\perp} = -e n_e \frac{\tilde{j}_\perp}{\sigma_\perp} - \frac{\textcolor{green}{3}}{2} \frac{e n_e^2}{\sigma_\perp \tilde{B}} \frac{1}{h_r} \frac{\partial T_e}{\partial u^r} \quad . \tag{25}$$

instead of:

$$\tilde{R}_{i,\perp} = +e n_e \frac{\tilde{j}_\perp}{\sigma_\perp} + \frac{\textcolor{red}{2}}{3} \frac{e n_e^2}{\sigma_\perp \tilde{B}} \frac{1}{h_r} \frac{\partial T_e}{\partial u^r} \quad . \tag{26}$$

- p.37, Eq. (2.103)

$$\begin{aligned}
& \frac{\partial}{\partial t} m_i n_i \tilde{V}_{i,\perp} + \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^\theta} \left(\frac{\sqrt{g}}{h_\theta} m_i n_i \tilde{V}_{i,\theta} \tilde{V}_{i,\perp} \right) + \\
& \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^r} \left(\frac{\sqrt{g}}{h_r} m_i n_i \tilde{V}_{i,r} \tilde{V}_{i,\perp} \right) + \beta_{div,\perp} + \chi_{div,\perp} + \\
& \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^\theta} \left[\frac{\sqrt{g}}{h_\theta} \eta_0 \frac{\tilde{B}_\phi}{\tilde{B}} \left(\frac{2}{3} \frac{\tilde{B}_\theta}{\tilde{B}} \frac{1}{h_\theta} \frac{\partial \tilde{V}_{i,\parallel}}{\partial u_\theta} - \right. \right. \\
& \left. \left. \frac{1}{3} \frac{\tilde{B}_\phi}{\tilde{B}} \frac{1}{h_\theta} \frac{\partial \tilde{V}_{i,\perp}}{\partial u^\theta} - \frac{1}{3} \frac{\textcolor{green}{1}}{\textcolor{green}{h_r}} \frac{\partial \tilde{V}_{i,r}}{\partial u^r} \right) \right] + \beta_{\Pi,\perp} + \chi_{\Pi,\perp} = \\
& - \frac{\tilde{B}_\phi}{\tilde{B}} \frac{1}{h_\theta} \frac{\partial p}{\partial u^\theta} + \tilde{j}_r \tilde{B} + \tilde{S}_{m_i \tilde{V}_{i,\perp}} \quad .
\end{aligned} \tag{27}$$

instead of:

$$\begin{aligned}
& \frac{\partial}{\partial t} m_i n_i \tilde{V}_{i,\perp} + \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^\theta} \left(\frac{\sqrt{g}}{h_\theta} m_i n_i \tilde{V}_{i,\theta} \tilde{V}_{i,\perp} \right) + \\
& \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^r} \left(\frac{\sqrt{g}}{h_r} m_i n_i \tilde{V}_{i,r} \tilde{V}_{i,\perp} \right) + \beta_{div,\perp} + \chi_{div,\perp} + \\
& \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^\theta} \left[\frac{\sqrt{g}}{h_\theta} \eta_0 \frac{\tilde{B}_\phi}{\tilde{B}} \left(\frac{2}{3} \frac{\tilde{B}_\theta}{\tilde{B}} \frac{1}{h_\theta} \frac{\partial \tilde{V}_{i,\parallel}}{\partial u_\theta} - \right. \right. \\
& \left. \left. \frac{1}{3} \frac{\tilde{B}_\phi}{\tilde{B}} \frac{1}{h_\theta} \frac{\partial \tilde{V}_{i,\perp}}{\partial u^\theta} - \frac{1}{3} \frac{\textcolor{red}{h_r}}{\partial u^r} \frac{\partial \tilde{V}_{i,r}}{\partial u^r} \right) \right] + \beta_{\Pi,\perp} + \chi_{\Pi,\perp} = \\
& - \frac{\tilde{B}_\phi}{\tilde{B}} \frac{1}{h_\theta} \frac{\partial p}{\partial u^\theta} + \tilde{j}_r \tilde{B} + \tilde{S}_{m_i \tilde{V}_{i,\perp}} \quad .
\end{aligned} \tag{28}$$

- p. 39, Eq. (2.107)

$$\begin{aligned}
& \frac{\partial}{\partial t} \left(\frac{3}{2} n_e T_e \right) + \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^\theta} \left\{ \frac{\sqrt{g}}{h_\theta} \frac{5}{2} T_e \left(n_e \tilde{V}_{i,\theta} - \frac{\tilde{j}_\theta}{e} \right) - \right. \\
& \frac{\sqrt{g}}{h_\theta} \left[\left(\frac{\tilde{B}_\theta^2}{\tilde{B}^2} \kappa_\parallel^e + \frac{\tilde{B}_\phi^2}{\tilde{B}^2} \kappa_\perp^e \right) \frac{1}{h_\theta} \frac{\partial T_e}{\partial u^\theta} - \frac{\tilde{B}_\phi}{\tilde{B}} \kappa_\wedge^e \frac{1}{h_r} \frac{\partial T_e}{\partial u^r} + \right. \\
& 0.71 \frac{\tilde{B}_\theta}{\tilde{B}} \frac{T_e}{e} \tilde{j}_\parallel + \frac{3}{2} \frac{\tilde{B}_\phi}{\tilde{B}} \frac{T_e}{e \omega_e \tau_e} \tilde{j}_r \left. \right\} + \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^r} \left[\frac{\sqrt{g}}{h_r} \frac{5}{2} T_e \left(n_e \tilde{V}_{i,r} - \frac{\tilde{j}_r}{e} \right) - \right. \\
& \frac{\sqrt{g}}{h_r} \left(\kappa_\perp^e \frac{1}{h_r} \frac{\partial T_e}{\partial u^r} + \frac{B_\phi}{B} \kappa_\wedge^e \frac{1}{h_\theta} \frac{\partial T_e}{\partial u^\theta} - \frac{3}{2} \frac{T_e}{e \omega_e \tau_e} j_\perp \right) \left. \right] = \\
& \left(\tilde{V}_{i,\theta} \frac{1}{h_\theta} \frac{\partial p_e}{\partial u^\theta} + \tilde{V}_{i,r} \frac{1}{h_r} \frac{\partial p_e}{\partial u^r} \right) - \tilde{j}_\theta \frac{1}{h_\theta} \frac{\partial \Phi}{\partial u^\theta} - \tilde{j}_r \frac{1}{h_r} \frac{\partial \Phi}{\partial u^r} \\
& + \tilde{j}_\perp \tilde{V}_{i,r} \tilde{B} - \tilde{j}_r \tilde{V}_{i,\perp} \tilde{B} + Q_{ei} + S_E^e \quad .
\end{aligned} \tag{29}$$

instead of:

$$\begin{aligned}
& \frac{\partial}{\partial t} \left(\frac{3}{2} n_e T_e \right) + \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^\theta} \left\{ \frac{\sqrt{g}}{h_\theta} \frac{5}{2} T_e \left(n_e \tilde{V}_{i,\theta} - \frac{\tilde{j}_\theta}{e} \right) - \right. \\
& \frac{\sqrt{g}}{h_\theta} \left[\left(\frac{\tilde{B}_\theta^2}{\tilde{B}^2} \kappa_\parallel^e + \frac{\tilde{B}_\phi^2}{\tilde{B}^2} \kappa_\perp^e \right) \frac{1}{h_\theta} \frac{\partial T_e}{\partial u^\theta} - \frac{\tilde{B}_\phi}{\tilde{B}} \kappa_\wedge^e \frac{1}{h_r} \frac{\partial T_e}{\partial u^r} + \right. \\
& 0.71 \frac{\tilde{B}_\theta}{\tilde{B}} \frac{T_e}{e} \tilde{j}_\parallel + \frac{3}{2} \frac{\tilde{B}_\phi}{\tilde{B}} \frac{T_e}{e \omega_e \tau_e} \tilde{j}_r \left. \right\} + \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^r} \left[\frac{\sqrt{g}}{h_r} \frac{5}{2} T_e \left(n_e \tilde{V}_{i,r} - \frac{\tilde{j}_r}{e} \right) - \right. \\
& \frac{\sqrt{g}}{h_r} \left(\kappa_\perp^e \frac{1}{h_r} \frac{\partial T_e}{\partial u^r} + \frac{B_\phi}{B} \kappa_\wedge^e \frac{1}{h_\theta} \frac{\partial T_e}{\partial u^\theta} - \frac{3}{2} \frac{T_e}{e \omega_e \tau_e} j_\perp \right) \left. \right] = \\
& \left(\tilde{V}_{i,\theta} \frac{1}{h_\theta} \frac{\partial p_e}{\partial u^\theta} + \tilde{V}_{i,r} \frac{1}{h_r} \frac{\partial p_e}{\partial u^r} \right) - \tilde{j}_\theta \frac{1}{h_\theta} \frac{\partial \Phi}{\partial u^\theta} - \tilde{j}_r \frac{1}{h_r} \frac{\partial \Phi}{\partial u^r} \\
& - \tilde{j}_\perp \tilde{V}_{i,r} \tilde{B} + \tilde{j}_r \tilde{V}_{i,\perp} \tilde{B} + Q_{ei} + S_E^e \quad .
\end{aligned} \tag{30}$$

- p. 107, Eq. (5.7)

$$\begin{aligned}
j_r = & \tag{31} \\
& \frac{1}{B} \left\{ \frac{1}{\sqrt{g}} \left[\frac{\partial}{\partial u^\theta} \left(\frac{\sqrt{g}}{h_\theta} m n V_\theta V_\perp \right) + \frac{\partial}{\partial u^r} \left(\frac{\sqrt{g}}{h_r} m n V_r V_\perp \right) \right] \right. \\
& \left. + \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^\theta} \left[\frac{\sqrt{g}}{h_\theta^2} \eta_0 \left(\frac{2}{3} \frac{B_\theta B_\phi}{B^2} \frac{\partial}{\partial u^\theta} V_\parallel - \frac{1}{3} \frac{B_\phi^2}{B^2} \frac{\partial}{\partial u^\theta} V_\perp \right) \right] - S_{m,\perp} \right\} \quad .
\end{aligned}$$

instead of :

$$\begin{aligned}
j_r = & \tag{32} \\
& \frac{1}{B} \left\{ \frac{1}{\sqrt{g}} \left[\frac{\partial}{\partial u^\theta} \left(\frac{\sqrt{g}}{h_\theta} m n V_\theta V_\perp \right) - \frac{\partial}{\partial u^r} \left(\frac{\sqrt{g}}{h_r} m n V_r V_\perp \right) \right] \right. \\
& \left. + \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^\theta} \left[\frac{\sqrt{g}}{h_\theta^2} \eta_0 \left(\frac{2}{3} \frac{B_\theta B_\phi}{B^2} \frac{\partial}{\partial u^\theta} V_\parallel - \frac{1}{3} \frac{B_\phi^2}{B^2} \frac{\partial}{\partial u^\theta} V_\perp \right) \right] - S_{m,\perp} \right\} \quad .
\end{aligned}$$

- p. 107, Eq. (5.7) In this equation the contribution of the pressure gradient is missing, because at that stage it was not encoded yet. Later applications do include this term (e.g. PSI, MITO 1994 and EPS, Montpellier 1994). The term arises at the right hand side as $+\frac{B_{\Phi}}{B^2 h_{\theta}} \frac{\partial p}{\partial \theta}$.